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Integration of ifc objects and facility management work information using Semantic Web

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ABSTRACT

The management of information throughout a building's lifecycle is becoming increasingly important, and building information modeling (BIM) is often used to ensure the interoperability of data. However, BIM-based facility information from the construction phase is difficult to access and use during the operation and maintenance phase. This occurs because the BIM information is not utilized well in facility management (FM). In this research, we propose an approach to effectively manage BIM-based FM information by linking the BIM-based building elements and FM work information in an FM system database. We present a Semantic Web-based FM information system that semantically links BIM data to relevant historical work records. The proposed ontology was evaluated using a sample dataset of the architectural maintenance work records of an office building. Using the proposed approach, facility managers will be able to increase their efficiency in searching related work records that consider shared BIM objects by enhancing the interoperability and accessibility of FM data via the Semantic Web.

1. Introduction

The concept of facility management (FM) is similar to that of building management, both of which are used to operate and maintain buildings. The planning of FM starts with the initiation of construction; then FM information is produced and managed throughout the building's lifecycle. Because FM can be affected by building information throughout the lifecycle, there is a great interest in the management of FM information and the object-oriented performance of FM works through a computerized information management system.

Data interoperability must be ensured through an object-oriented work process that covers the entire lifecycle of the building. To use this approach, especially in the FM area, a standard specification or type of data should be defined for a given facility management system (FMS). To use advanced information management, building information modeling (BIM) ensures coordinated, consistent, and computable building information in the architectural, construction, and engineering (AEC) industry by buildingSMART International [1]. Thus, the schema of industry foundation classes (IFC[1]) was developed as an integrated object data model to systematically and semantically manage a building's elements throughout its lifecycle. Since the IFC schema has semantic data relationships among the building data entities [2], the building data generated and managed in each phase of the lifecycle can be linked not to the internal data entities but to the external heterogeneous database (DB) of the information management system, such as material specifications or actual costs of construction activities.

Although information related to the design and construction of a building could be managed by an international standard, FMS data in the FM area cannot be managed by a public data model because the FMS is used differently by the various FMS scopes and functions in the operation and maintenance (O&M) phase. To define the specification of the data exchange for FMS at the start of the O&M phase, a global development of exchange requirements has been developed for the handover procedure, known as the 'construction operation building information exchange' (COBie) system. The COBie schema can be used as a standard data exchange system throughout the building's lifecycle to ensure the interoperability of BIM data for a FMS using a data spreadsheet. However, when the IFC-based BIM data are used with COBie for an FM work, the required FMS data should be managed by an integrated, IFC-based building model with some external data, such as the FM work information, which are managed separately in the FMS DB. Additionally, in the real world, because the data structure of an

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[1] http://www.buildingsmart-tech.org/ifc/IFC2x4/rc4/html/index.htm

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FMS database is not normalized to link the IFC objects or the COBie data as a public data model, the facility manager rarely uses an FMS on the actual job site via the traditional approach without IFC objects or COBie [3]. To increase the usability and accessibility of the object-oriented BIM data in an FMS, the building data, which are generated separately by a heterogeneous data schema, should be integrated semantically to provide object-based FM information that supports BIM-based FM works, such as 3D visualization, review of related work records, and maintenance work management.

Therefore, in this paper, we propose an advanced data management approach to integrate IFC objects, COBie data, and maintenance work information of the FMS DB. In our approach, an ontological system is developed through the Semantic Web to enhance the usability and accessibility of object-oriented information management with FMS. To that end, after analyzing the technical developments of information management in the FM area with BIM and the Semantic Web, we developed an ontology to link the IFC objects with the FM work information of a traditional FMS DB. To evaluate the proposed approach, we collected the historical maintenance work records from a FM company, and used the proposed ontology to semantically link the IFC-based, common BIM files via buildingSMART to the maintenance work records.

2. Technical improvements of information management for FM

The increased use of information technology (IT) and related applications has provided advantages such as improved collection of the data required for FM works and the simplification of the FM work process [4]. According to Jawadekar [5], the development of information management in the FM area was improved with specific IT. As a first step, the computerized information management system could be used for storing FM information, automating data processes, and conducting comprehensive reviews of the physical and functional properties of a facility [6]. The second most important improvement to FM by IT applications is the real-time exchange of data via Web-based information management [7]. Thus, the quality rather than the quantity of FM information became the focus of the facility manager's attention.

To increase the quality of FM information, including BIM data, the FMS must be well developed with precise and valuable data collected by all stakeholders in the previous phases. A variety of approaches can be used to support the FMS by importing and exporting data, such as text, spreadsheets, and relational DB files. Moreover, heterogeneous data are collected by these different methodologies [8]. However, the development of an FMS database still involves some potential problems [9]: 1) the stakeholder's roles and responsibilities for FMS requirements are often unclear; 2) because the applicability and reliability of the data inputted to the FMS DB are often insufficient, the FMS manager are required to repeat previous work in order to develop a FMS DB; and 3) because the requirements of FMS are managed manually based on 2D building information, human error may be introduced to the relationships between FMS requirements.

2.1. BIM-based information management

Various commercial FMS systems use the FM information management tool for BIM. The development of FMS to search for and provide accurate FM information is required in order to increase the type or amount of FM information available to support FM works [10]. To use FMS for efficient information management, the building information entered into the FMS should be organized by semantic relationships among the component data to integrate heterogeneous information. In general, building information has limited heterogeneity because various stakeholders use different software to manage it, but the heterogeneity can be reduced by completing data interoperability with semantic relations [11].

For this reason, a significant amount of research and development has been conducted to ensure the interoperability of FM information using BIM. Kivits and Furneaux [12] analyzed the availability of information from BIM-based FM case studies, and suggested that FM information should be provided more easily and rapidly to facility managers. In addition, FM information should be updated to reflect current status within a single, integrated BIM model in the O&M phase.

McArthur's research [13] identifies improvements in four major areas that are necessary to use BIM in FM works: 1) identification of critical information required to guide operational decisions; 2) high level of effort to create new or modify existing BIM models; 3) management of information transfer between real-time operations and monitoring systems and the BIM model; and 4) the handling of uncertainty based on incomplete building documentation. These four improvements enhance the possibility of using a BIM model in the O&M phase, and the accuracy and consistency of the FM information was verified by three case studies using BIM.

Kang and Choi [14] suggested a BIM-based work scenario in Korean FM practices to support general maintenance works. To validate the proposed scenario, they interviewed a facility manager who had more than 15 years of experience in the management of FM information using BIM, and the manager indicated that the productivity of maintenance work had increased from 25% to as much as 205% as a result of several improvements in requesting, searching, verifying, reviewing, and collecting the required information.

When BIM is used in the design and construction phase of a new construction project, a COBie system may exchange data between phases of a building's lifecycle. COBie is an international standard for the exchange of building information from the design phase to the O&M phase, using a formal spreadsheet [15]. However, when using the COBie sheet for data exchange, the required data should be managed by the integrated, BIM model with some external data. When information managers want to collect FM data using COBie, they must consider external data, such as product data, using a specifier's properties information exchange (SPie) library, which uses buildingSMART as middleware to apply the SPie library to the COBie sheet [16].

2.2. Integrated management using Semantic Web

When automating the process of building information through a computerized DB, two conditions are necessary for semantically-related data interoperability. First, the information must be represented by computerized codes in a programming language. Second, ontology is required to define the relationships among concepts of building information-coded, computerized language [17]. After the development of the World Wide Web (WWW) in the early 1990s, Tim Berners-Lee proposed an extension—the Semantic Web—which can handle Web data without human intervention [18]. Because Web data are currently expressed in human-readable form, computers cannot interpret their exact meaning. To overcome this, Semantic Web data are expressed in a computer-readable format using ontologies, the key technology of the Semantic Web, which are formal and consensual specifications of conceptualizations that provide a shared understanding of a certain domain [19]. Using the ontology that describes various concepts and their relationships, the computer automatically enables reasoning about data from different data sources in order to manipulate relevant data or find new information.

To use BIM-based data relationships in a semantic inference procedure, researchers have converted IFC files to Web ontology language (OWL) for input into Semantic Web applications. Because a BIM model using an IFC is represented in building information as computerized EXPRESS language, IFC-based building information can be defined by OWL-based data and relations with the similarity of the data relationship between IFC and OWL [20]. However, because the IFC schema is described by logic-based EXPRESS, the semantic relationship is limited to the representation and sharing of inferred knowledge. Therefore, awareness is necessary for linking additional semantic information of



Fig. 1. Technical milestones of information management in the FM area.

an IFC-based BIM model and to manage inferred knowledge with semantic relationships. Barbau suggested the linked data model for product information to integrate geometry data using the IFC format with non-geometry data from other computerized DB using ontology [21]. This contributed the semantic link between the building information and external product properties in DB using ontology. In addition, after IFC-based building elements had been converted into instances via the resource description framework (RDF) structure, the inferred knowledge can be recognized by an OWL-based semantic inference process using the proposed application [22].

BIM information represented by IFC is automatically extracted to constitute an ontology for the transformation of data relationships between IFC and the ontology [23]. Thus, the relationships extracted from IFC can be re-used directly for semantic inference in the OWL-based ontology. Various research efforts have explored IFC data for automatic extraction and systematic conversion to OWL, which is referred to as 'IFC to RDF' [24], aiming to ensure that the BIM-based information has the expandability to be used more extensively in the design area and other related areas [25].

IT for FM falls into three areas: 1) information management in FM; 2) development of BIM; and 3) Semantic Web-based information management. Fig. 1 shows several major milestones in these three aspects of IT development. From the practical perspective, several commercial BIM-based FMSs have been provided by vendors who are developing BIM-based applications to support the design and construction process; thus, BIM-based data management is not yet capable of supporting real-world FM practice, with the exception of some exemplary cases from academia.

3. Ontology for integration of FM information

3.1. Overview of ontology

1) Definition of FM ontology

In general, the main BIM-based information consists of a COBie data sheet and an IFC-based BIM model, including attached documents to represent detailed or additional contents of the geometric and non-geometric information. Because the COBie and IFC have different schema, the data relationships in the O&M phase using BIM-based FMS are organized separately. In this regard, even the IFC schema can be used to convert to a COBie data sheet; thus there is one way to extract IFC data in order to input specific data components of the COBie system. Accordingly, when the IFC is updated, a facility manager must modify it using a BIM authoring tool, such as Revit[2] or ArchiCAD[3] software. However, because the BIM environment is used to access or manage the BIM data in the O&M phase, an IFC-based BIM model has difficulty referencing the FM practice with a BIM-based FMS. In addition, a COBie data sheet must have additional IFC data to support the input data of an FMS, including the types, systems, and job information of a facility. Thus, middleware has been developed to input the additional information of COBie into the BIM authoring tool as an add-on application, such as the COBie Toolkit by Autodesk.[4]

Because extensive information is generated in various areas of the integrated IFC file during a building's lifecycle, a facility manager must define whether the required FM information from the IFC file is mandatory or optional for each FM work. Since the facility manager cannot recognize all of the required FM information, buildingSMART suggested in 2009 the use of the IFC-based model view definition (MVD) to ensure the handover to FM. However, because the MVD described the scope of the IFC schema, the heterogeneous FM information required from the other applications is not used to support FM works.

Therefore, to integrate the FM information from various DB structures and automatically define the data relationships among FM information, ontology is necessary to computerize the knowledge and semantically relate the data interoperability as middleware of BIM-based FMSs. The ontology of information management can reduce redundant data input, omitted data, and data errors through the sharing of knowledge and re-use of information [26].

In this study, we developed an OWL-based ontology to manage the required FM information from BIM-based IFC and COBie data. The developed ontology integrates the knowledge systems of FM information as a schema, and the inferred knowledge can be processed by computer without human error. Thus, because the IFC schema considers the semantic relationships among data, the data relationships of

[2] http://www.autodesk.com/products/revit-family/overview.
[3] http://www.graphisoft.com/archicad/.
[4] http://www.biminteroperabilitytools.com/cobieextension/help/withrevit.html.

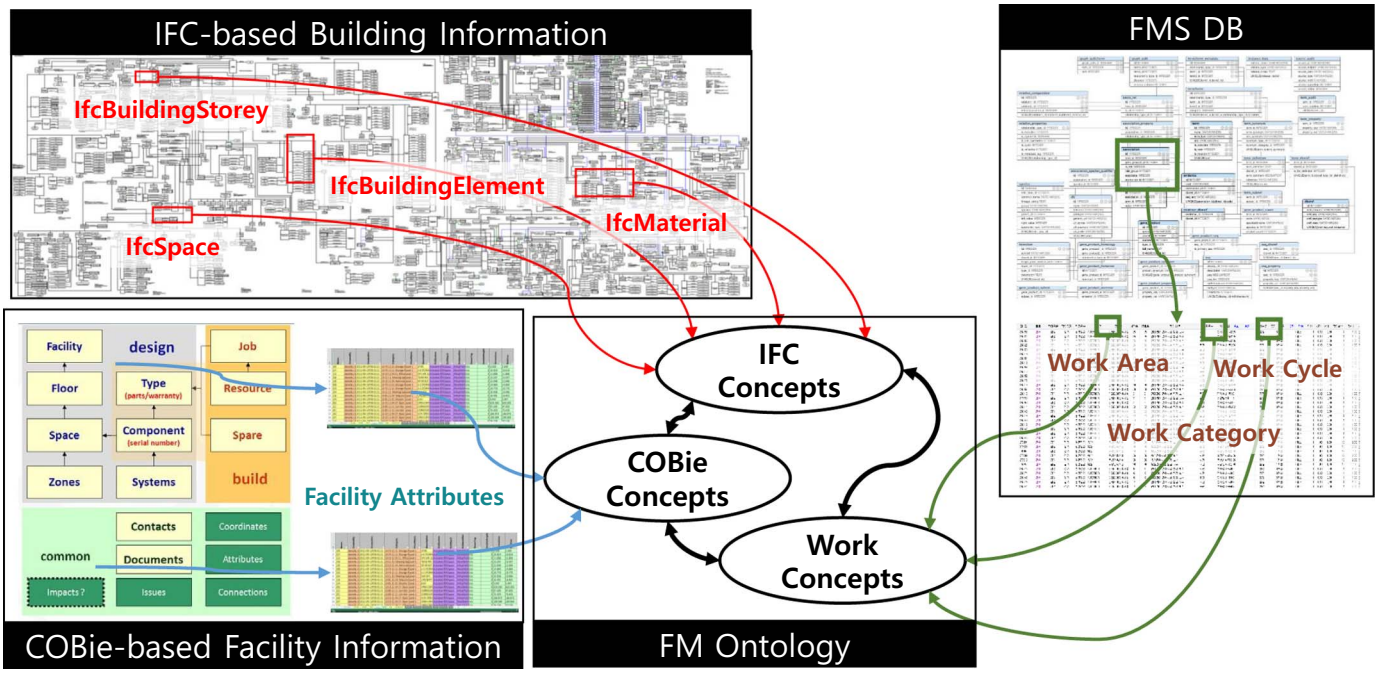


Fig. 2. Conceptual overview of FM ontology.

IFC can be transformed directly to develop the relations of the OWL-based ontology. With the proposed ontology, the Semantic Web can use the IFC concepts as building elements information from an IFC schema, COBie concepts as a facility's attributes from the COBie data sheet, and work concepts as FM work information from an FMS DB. Fig. 2 shows the relationships among these concepts.

2) Conceptualization

In this study, the proposed ontology provides the BIM data environment to incorporate BIM into FMS through BIM-based FM information, which semantically links to the FM works information in an FMS DB. Using the proposed ontology, a facility manager without experience or skill in BIM can search for the required BIM-based FM information, such as the elements of the FM work scope and its properties. In addition, the FM organization can manage the FM information efficiently by automatic linkage between BIM and the traditional FMS DB without customizing the FMS as BIM. The proposed ontology integrates the IFC-based BIM data from the design and construction phases with the FMS DB. In the O&M phase, the as-built geometry and information concerning the facility's attributes are integrated with the FM work information via the Semantic Web.

To integrate the FM information, the ontology defines several concepts—namely, the type of work, facility information, and historical work records. Thus, the concept of work type is comprised of work area, work cycle, and work category to recognize the type of FM works from a traditional FMS DB. The concept of facility information is comprised of relative space, element, and attribute information in the scope of the FM works. Relevant space and element information can be extracted from IFC-based BIM data, and the attributes can be converted from a COBie data sheet to the ontology (Fig. 3).

Because the COBie data sheet can be generated from the IFC file, there are several equivalent concepts between IFC and COBie. For instance, the information of IfcSpace in IFC describes the same as a space sheet in COBie, even though the data components differ. In this case, the space name, Omniclass classification or the person who manages the space information can be found in the COBie space sheet, but the related space boundary, windows, and assets can be found in the IFC file using the IFC schema. Therefore, the classes, which describe equivalent concepts in IFC and COBie, are related by equivalent and object property to develop the proposed ontology. An equivalent relationship means the concepts of classes are exactly the same, and an object property means that a relationship exists between the classes. There could be one of many relationships, such as hasSpace, hasBuildinfElement, and others. Fig. 4 shows the semantic relationships of the classes by the concepts.

3.2. Process of using FM ontology

With the proposed ontology, the BIM-based FM information generated in the construction phase can be used directly for FMS in the O&M phase as FM information integrated via the Semantic Web. Moreover, the historical FM work records related to the subject FM work can be managed using BIM through the related building elements and spaces. To that end, the required IFC and COBie data are extracted from the BIM-based facility information then, converted to RDF data to bind with the OWL-based ontology. In the FM ontology, the historical work records and FM work data are used as inputs to the ontology as individual instances in order to define the type of FM works. Then, for semantic data mapping, the RDF data and OWL-based ontology are bound to each other as integrated data. When the data are bound, the semantic inference engine provides the inferred knowledge in XML format. In the inferred knowledge, the classes of IFC and COBie are related semantically with the FM information, including related space and element information and related historical FM work information to reference the current FM work. With relative space and element information, a facility manager can review the work scope by IFC object using the 3D-based building geometry of an IFC-based BIM model (Fig. 5).

To describe the use of integrated FM information as a system architecture, six layers are needed to obtain the inferred knowledge from the integrated FM information: 1) the IFC data extraction layer; 2) COBie data exchange layer; 3) FMS data management layer; 4) ontology management layer; 5) reasoning layer; and 6) query layer (Fig. 6).

1) IFC data extraction layer

IFC-based spatial classes in the proposed ontology can be extracted automatically and systematically from the IFC file. To extract IFC

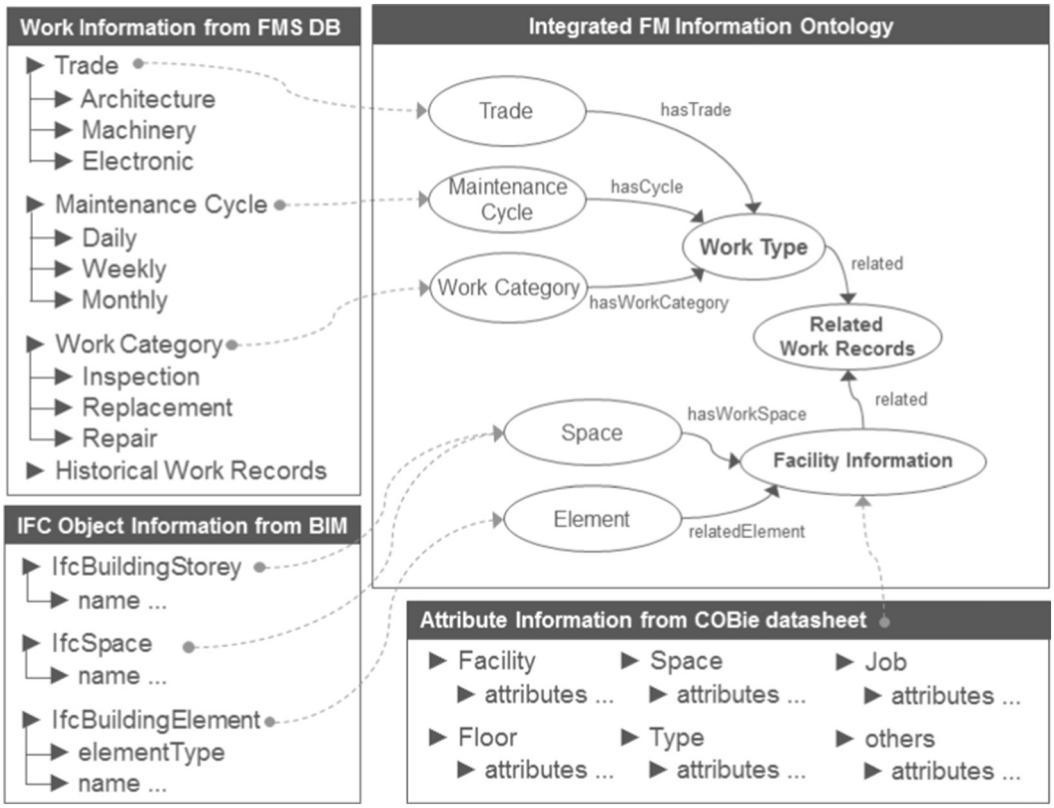


Fig. 3. Composition of FM ontology.

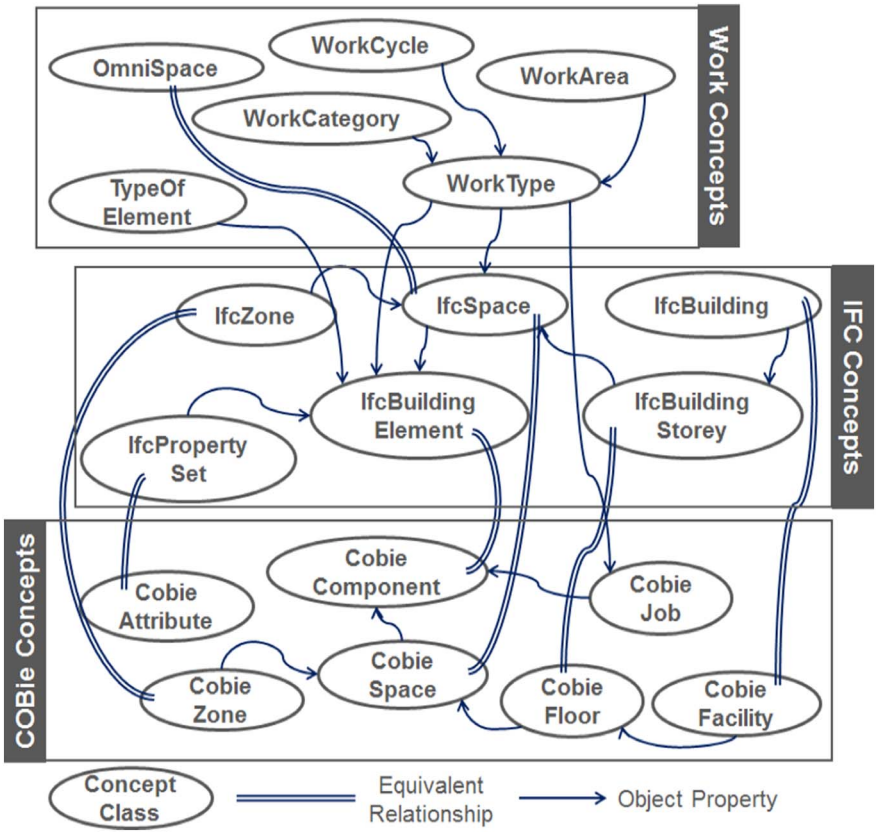


Fig. 4. Semantic relationships of the classes by the concepts.

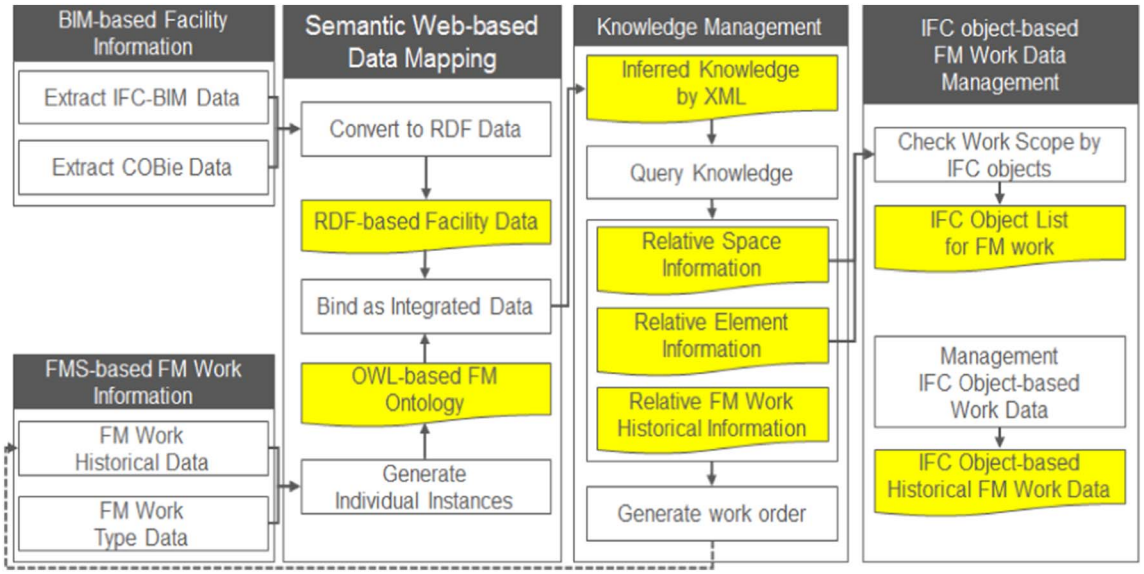


Fig. 5. Data flow of integrated FM information.

spatial information, the IfcSpace entity has to be recognized for storing the value of ID property. In this study, because the project, site, and building information of the spatial information can be extracted by the same algorithm as the IfcSpace entity, various names of properties can be stored into the object information DB. Then, the type of recognized IfcSpace entity is stored by 'IfcSpace' in order to distinguish from other types of extracted entities. After the object type has been stored, the properties are extracted and stored by the IFC schema: name;

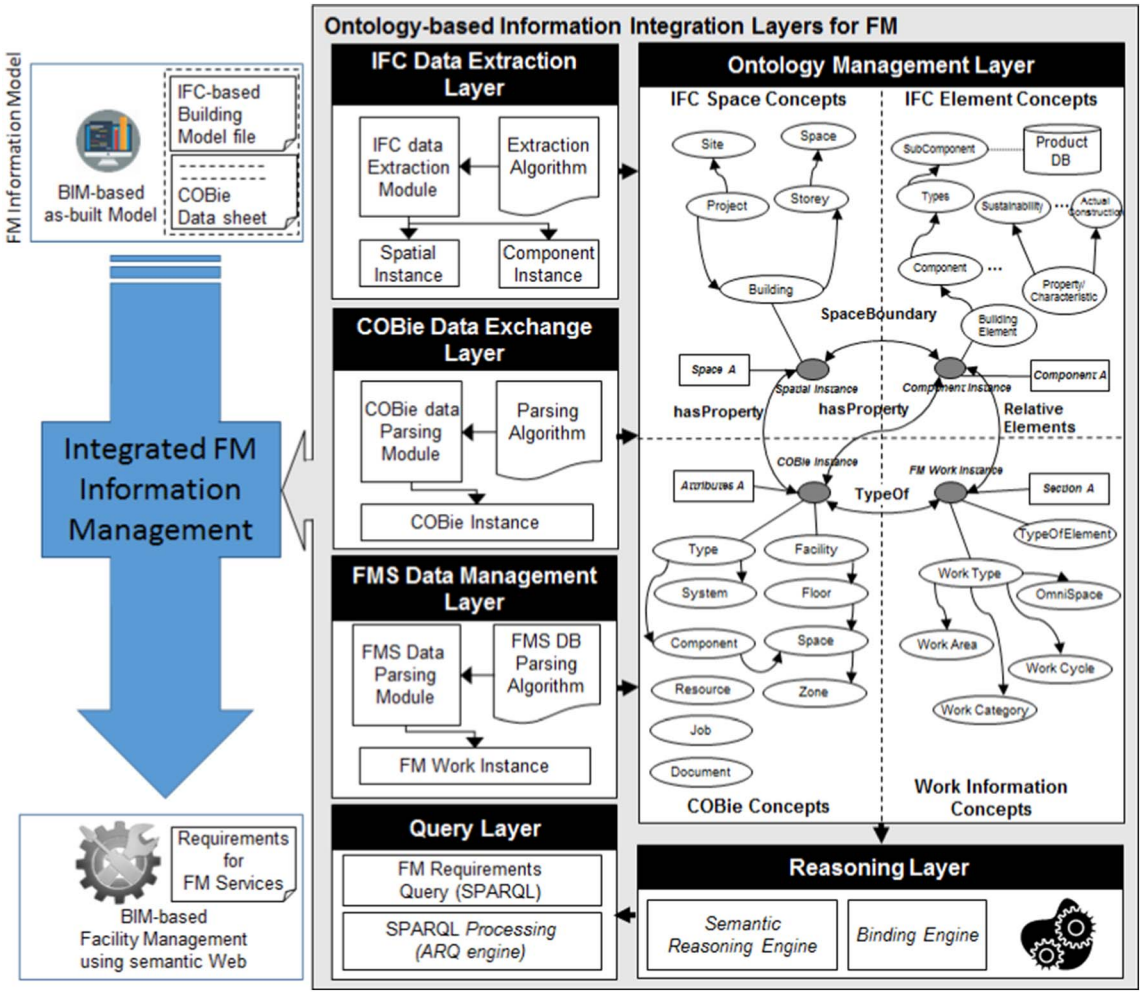


Fig. 6. Integrated FM information management process.

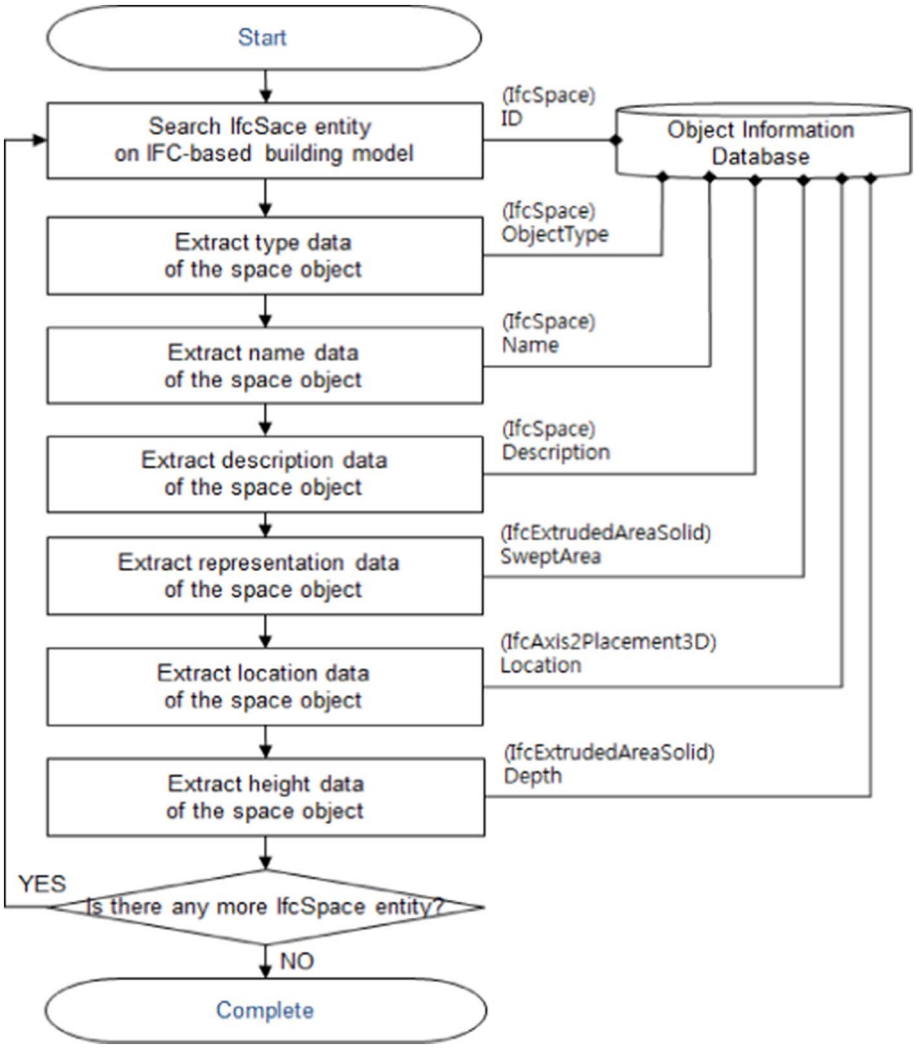


Fig. 7. Algorithm used to extract IFC spatial objects.

description, which generally has the value of the usage in practice; area; and location data into the object information DB. Fig. 7 shows the data extraction process of the IfcSpace entity.

To extract information of related building elements, the space boundary elements are determined for the recognized space. The building elements, which are constructed as space boundaries, are extracted to match the scope of the FM works because there are too many building elements in the IFC file (e.g., furniture, walls, columns) for an interior design object. Thus, the IfcRelSpaceBoundary entity is found to search the valid building element referenced by the recognized IfcSpace entity as a value of the relatingSpace property. After searching the IfcRelspaceBoundary, the ID of the relatedElements as IfcBuildinfElement is used to extract data. Then, several properties of the IfcBuildinfElement entity (e.g., name, objectType, representation data from the IFC schema) are stored into the object information DB. This allows the material information to be extracted by the IfcMateriallayerSetUsage entity by referencing the recognized IfcBuildingElement entity (Fig. 8).

2) COBie data exchange layer

Because all of the data on the COBie data-sheet are not mandatory, authors frequently omit COBie data from the previous phase. In such cases, the COBie data sheet must be used for other facility information from heterogeneous DB, including the IFC file or other electronic documents for the data-based FM works. As the most important issue is the conversion of the COBie data, the units and type of COBie data must be ascertained in order to ensure consistency with the other FM information. Because the COBie data sheet is formatted by XML for direct use with an FMS DB, the results cannot be used in FM practice if the units of the area, length, and volume differ from those of the other FM information. In addition, the type of data—string, integer, or binary (Boolean) types—ensures the values' accuracy.

3) FMS data management layer

Because inputting the work information into an FMS may be tedious and repetitive for a facility manager, the accuracy and efficiency of the work scope and contents dramatically decrease. In this regard, the scope and contents of the FM work consist of various objects, so the information can be inputted in the abstract or omitted, and searches may provide poor results.

In this study, the concept of work information in an FMS DB was semantically defined by the work area, work cycle, and work category in order to link with the historical FM work records. Here, the type of FM work can be used to identify related historical work records beyond the keywords; the results contain the various FM works even if they were conducted with a different scope and contents. To define the classes of FM work information, additional data components are required to link with the other FM information, such as related spaces and elements. The related spaces and elements have multiple values to represent the work scope, and the value of the property has to be linked

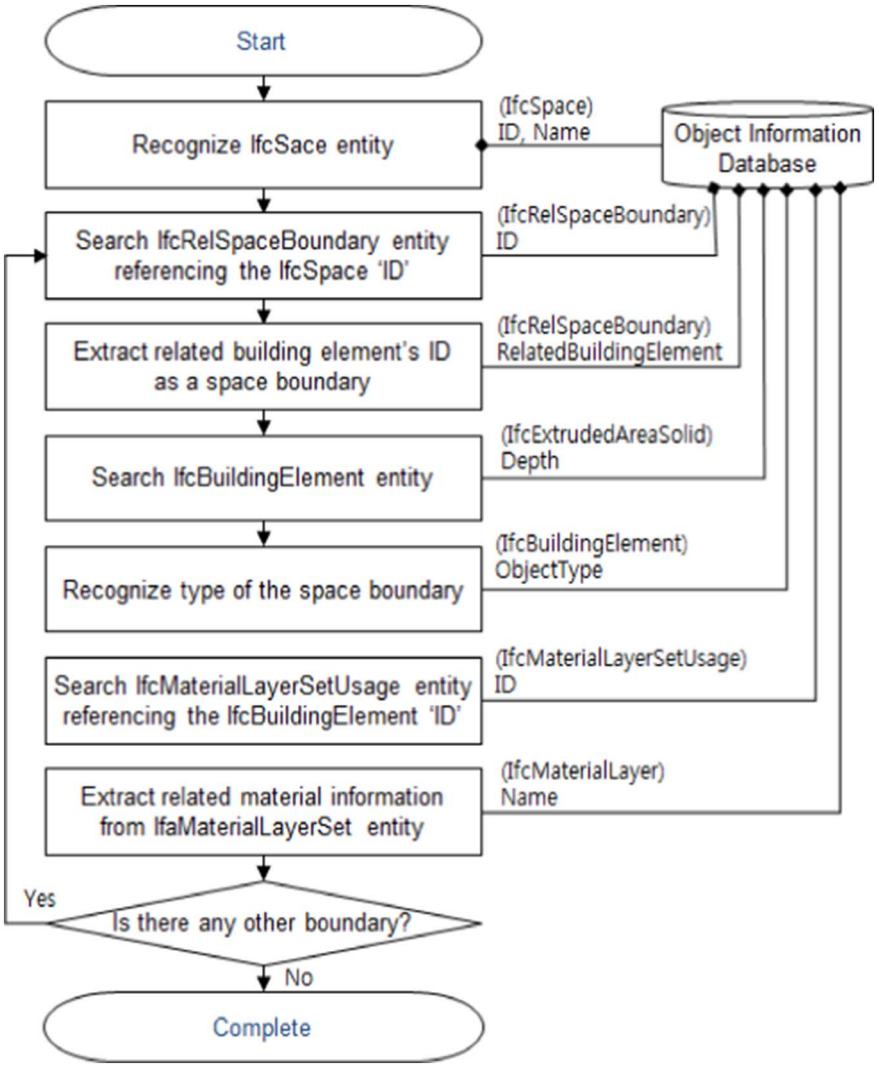


Fig. 8. Algorithm for the extraction of IFC building elements.

with the IFC objects. To do so, the building elements, which are composed of the IfcSpaceBoundary, can be used to automate the input of the work scope as the IFC objects.

4) Ontology management layer

An ontology management layer manages the class, instance, and property of the concepts for FM information constituted by the proposed ontology based on OWL format. In this regard, the classes and property, including the object and datatype properties, should ensure the ability to explain the knowledge system for FM, and when a new concept is needed to support the FM works, the original concepts should have expandability to link with the new class and properties via a global standard format, such as OWL or RDF.

In this study, the FM ontology provides the semantic linkage of BIM-based FM information and FM works information by mapping the IFC objects and historical FM works records; thus, new concepts can be added to the proposed ontology in order to support the various objectives and the functions of FMS. In this case, an interface is necessary to merge the ontology as an equivalent relationship or object property between the original and new concepts.

5) Reasoning layer

The reasoning layer binds the integrated data by RDF format and the ontology by OWL format, including the work information as an instance, and it also reasoning semantically the bound file by the reasoning layer. In this regard, binding the RDF and OWL files occurs in the new classes or instances for the FM ontology from the expandability perspective. From the ontology, a classification of class hierarchy is used to infer the knowledge semantically with the new classes or the instances from the RDF files. To bind the RDF and OWL files, a JAVA-based binding engine is usually used such as open source Jena API. Then, there are various reasoning engines in the computer science area including commercial engines and open source engines like BOSSAM API. We used the BOSSAM reasoner, which has an RETE-based rule engine with native support, for reasoning over RDF and OWL (http://mknows.etri.re.kr/bossam/). The overall process for creating the inferred knowledge consists of three steps:

- Step 1: The reasoning layer binds the two domain ontologies and material RDF data via the Jena API into bound knowledge, which is a simple, non-redundant collection of ontologies and material RDF data;
- Step 2: The BOSSAM reasoner takes the bound knowledge and generates inferred knowledge using its sub-reasoners (e.g., the OWL DL reasoner); and
- Step 3: The knowledge base stores the inferred knowledge.

6) Query layer

To retrieve the data required for FM works, a computerized query

Table 1
Comparison of methods for FM works using FMS.

FM company	Access of FM information	Usage of FMS	Work records management
A company	2D-based paper documents	Only used to electronic approvals	Based on paper documents
B company	2D-based electronic documents	Limited used to work records management	Based on paper documents and work codes are managed on FMS
C company	2D-based electronic documents	Manual data input on routine and occasional FM works	Using internal standard with FMS DB
D company	2D-based electronic documents	Manual data input on routine and occasional FM works	Using internal standard with FMS DB
E company	2D-based electronic documents	Manual data input on routine and occasional FM works	Using internal standard with FMS DB
Proposed ontology	IFC-based BIM data and COBie data	BIM data automatically links to FM works	Mapping IFC objects and FM works

Table 2
Examples of maintenance work records.

Date	Area	Elements	Work contents	Labors	Work time (h)	Working floor
20160115	Finishing	Sealant	Putty works damaged part of wall in meeting room	2	2	25
20160503	Finishing	Architectural finishing	Check and repair the fragile part of water leak in a building	3	4.3	7
20160603	Finishing	Gypsum board	Putty works of wall in executive suite	2	3	9
20160530	Curtain	Curtain	Repair the line of blind in meeting room	4	4	31
20160406	Door	Access door	Check and repair the access door in each room	2	3	−1
20160310	Door	Door & lock-set	Clean the door and lock-set in each room	4	12	10–35
20160118	Stone	Stone	Repair the wobble stone at the front of main gate	2	3	1
20160405	Tile	Ceiling tile	Pollution check and repair the ceiling tiles in meeting room	3	6	16
20160202	Tile	Carpet tile	Pollution part of the Carpet in lounge	2	10	15–26
20160127	Tile	Other tile	Bulging treatment of the floor tiles in cargo bay	2	2	31
20160601	Architecture	Common	Condition check the door in each room	1	1.3	−3–1
20160322	Architecture	Common	Attach the ceramic non-slip board in expansion joint	3	3	−2
20160502	Management facility	Management facility	Epoxy coating the floor in generator room	3	6	−3
20160525	Paint	Water paint	Check and sealing the pollution part of columns in meeting room	3	6	15–18
20160318	Paint	Oil paint	Sealing the door in situation room	3	12	−1
20160510	Toilet	Toilet architecture	Check and repair the utilities in toilet	2	10.6	−3–35

can be used to map the IFC objects and FM work information queries. To support the user interface for the query, an application can be developed endpoint based on various query engines that support OWL-based ontology, such as the ARQ engine (http://jena.sourceforge.net/ARQ) using Jena API. When an expert inputs FM work information queries via a query endpoint application, the query layer submits them to the ARQ engine, which then retrieves relevant IFC objects from the knowledge base. Query results are returned in XML, which is easy to use in an FMS. In these results, there is FM information, such as work types, IFC objects (e.g., spaces and building elements), and facility properties from COBie.

4. Evaluation

4.1. Overview

To validate the proposed ontology, we conducted a case study in three aspects of a practical situation: maintenance, operation, and FM work management. To that end, we collected data from five Korean FM companies that conducted the outsourcing of FM. In general, most buildings were operated in one of two ways—direct management or outsourcing to an external company, especially FM works in Korea. In this case, the FM work level of the direct management type is usually lower than the outsourcing type because there is a legal standard related to FM work within the safety area and other specific areas but not maintenance work in regards to FM quality. Because it depends on the organization's responsibility to manage a facility, the quality and efficiency are dramatically decreased. Moreover, an IFC-based BIM model cannot support current FM works in practice, but the benefits of BIM use in the O&M phase have only been proven in academic studies.

In this study, we compared the FM methods of five companies using FMS without BIM. Table 1 shows a comparison of the methods of FM works. According to this table, most of the FM companies used an FMS, but the scope of FMS usage varied in regards to 2D-based electronic documents. Companies A and B used the FMS only to manage the work records or electronic approval of the FM works. The other companies used the FMS in the interest of efficient data management to help FM works. However, the companies inputted their work information manually with the work scope and the contents as an abstract.

FM companies have an internal work standard that they use to classify the FM work by work type, work cycle, and work category because the FMS is customized to the organization by that standard. Usually, FM works are divided into the categories of routine and occasional work. In the case of routine work, the work is managed using FMS by the internal standard with an annual FM strategy, including work contents and work cycle. In the case of occasional work, the work is categorized by area, such as architecture, machinery, electronics, and disaster preventions. Occasional FM work is much more frequent than routine FM work; for example, Company E had almost 600 records of a building in which occasional maintenance work had been conducted in the architectural area over just about five months (1/01/2016 to 6/02/2016).

Table 2 shows examples of the maintenance work records, which translated the work contents into English. In Table 2, a person in charge of the maintenance work manually entered the work information, including work category, subjective elements, work contents, number of laborers, working time, and the locations of the floors into the FMS DB. Using the FMS DB, however, when a facility manager was seeking a specific building element to do maintenance work, there was insufficient facility information to review previous maintenance work,

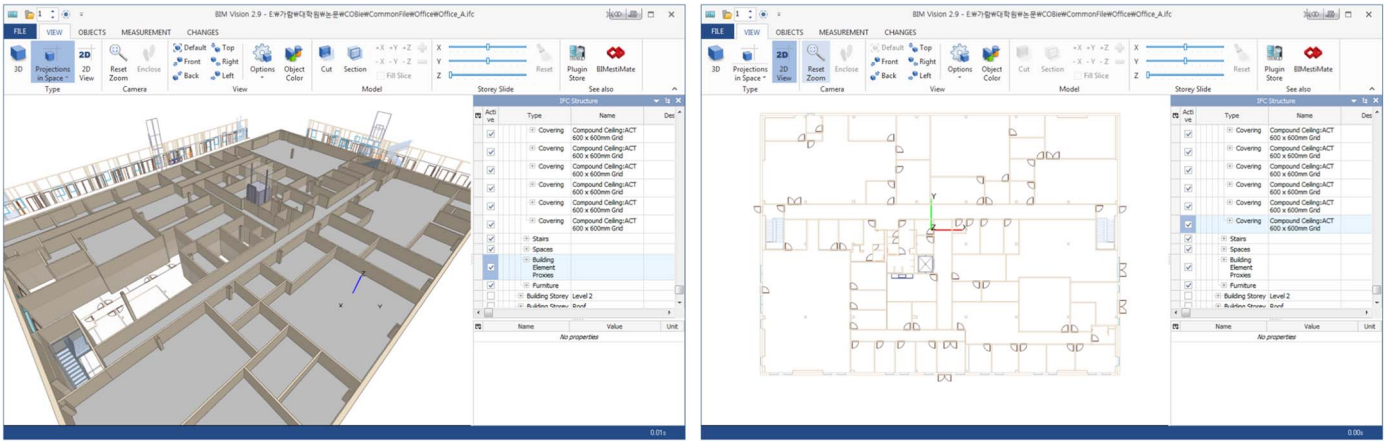


Fig. 9. Common BIM file-office by buildingSMART.

Table 3
Extracted IFC spatial concepts.

Concept	IFC entity	IFC ID	Attribute	Value
Building	IfcBuilding	#88	Name	Office
Building story	IfcBuildingStorey	#103	Name	Level 1
Space in 'Level 1'	IfcSpace	#135	Name	101
	IfcSpace	#311	Name	102
	IfcSpace	#482	Name	103
	IfcSpace	#645	Name	104
	IfcSpace	#807	Name	105
	IfcSpace	#991	Name	106
	IfcSpace	#1235	Name	107
	IfcSpace	#1466	Name	108
	IfcSpace	#1686	Name	109
	IfcSpace	#1913	Name	110
	IfcSpace	#2173	Name	111
	… about 60 spaces			

because the records lacked data on the specific object. In addition, if maintenance work has been conducted in multiple spaces, such as a zone or district, the manager cannot determine the space or even the work that was done because there was no keyword for the name of the space.

To evaluate the proposed approach, we conducted a case study using 590 records of occasional maintenance work in an architectural area with a BIM-based sample model to validate the proposed ontology pursuing the linkage of IFC objects and maintenance work records. The IFC-based BIM model was adapted to the 'common BIM file – office' developed by buildingSMART to ensure its quality.[5] The sample IFC file has two stories—Level 1 and Level 2. Level 1 has 60 spaces, 262 walls, 66 doors, 25 windows, two stairs, and an elevator with a net area of 1814.786 interior square meters, including the COBie data-sheet (Fig. 9).

4.2. Implementation

1) Data extraction

The sample IFC file contains about 6487 k bytes of facility information including IFC objects. As we mentioned earlier, to extract the required information, we recognized the data entities of the IFC spaces and building elements according to the IFC data extraction layer focused on Level 1. Tables 3 and 4 show some of the extracted IFC data using the proposed extraction process. In Table 3, the concepts of the building, the building story, and the space are extracted with object type, ID, and attribute names. Table 4 shows the concept of building elements, including IfcSpaceBoundary, IfcCovering, IfcMaterialLayerSet, IfcWallStandardcase, and IfcDoor, with object type, ID, and attributes of the relatedID to reference the IfcBuildinfElement, relatedType as an object type of a related element, relatedName as the name of a building element, and finMaterial as the finishing material of a building element.

2) Data exchange

Meanwhile, to transfer the COBie data from an XML to RDF structure, the contact, facility, floor, space, and zone sheets are translated into RDF data. The contact sheet indicates the person in charge of the generation in regards to the COBie data sheet with 14 attributes of personal information. The facility sheet contains the building information with nine attributes of general information and five measurement units. The floor sheet contains the building story information with seven attributes. The space sheet contains 60 space objects with nine attributes including area and height. The zone sheet contains the district information with six attributes. After converting to RDF data, the OWL-based ontology file has bound the RDF file using JAVA-based Jena 2.6.4 API. In this regard, the output of the Jena 2.6.4 API is formatted by OWL, which contains the IFC spatial and building element concepts and COBie data by RDF triple structure.

3) Ontology development

The OWL-based ontology, on the other hand, was developed for this case study to link concepts among the BIM-based FM information. To develop the ontology, the maintenance work records need to be coded into computerized classes and relationships using the Protégé application. In this case study, we used the Protégé 3.4.4 to define the ontology by OWL within the maintenance work records, IFC objects and COBie data. Fig. 10 shows the object properties of the classes in the developed ontology for this case study. The FM information is defined by three types of concepts—Work Information, IFC Objects, and COBie Concepts. The classes, understandably, are composed of a hierarchy to define the concept and relationship. For instance, since a maintenance record has the work type listed as repair and the building element listed as a wall, the work information can be automatically linked with the specific wall object of the IFC objects. Relationships of spatial objects in the ontology are based on the IFC schema, while the building story and zone could be defined by an IFC object. Moreover, the related COBie data-sheet of the IFC-based BIM model can be also linked with the object properties of the ontology. Consequently, after the inferred knowledge is generated by BOSSAM reasoning engine through XML, semantic relationships are identified among the concepts of IFC, COBie and work information.

[5] https://www.nibs.org/?page=bsa_commonbimfiles.

Table 4
Extracted IFC building element concept.

Concept	IFC entity	IFC ID	Attribute	Value
Building element	IfcSpaceBoundary	#160910	relatedID	#84167
	IfcCovering	#84167	relatedType	ceiling
	IfcCovering	#84167	relatedName	Compound Ceiling:ACT 600 × 600 mm Grid:198,077
	IfcMaterialayerSet	#80352	finMaterial	Ceiling Tile 600 × 600
	IfcSpaceBoundary	#160912	relatedID	#32465
	IfcWallStandardcase	#32465	relatedType	wall
	IfcWallStandardcase	#32465	relatedName	Basic Wall:Interior - Partition (92 mm Stud):149,176
	IfcMaterialayerSet	#26383	finMaterial	Plasterboard
	IfcSpaceBoundary	#160913	relatedID	#63713
	IfcDoor	#63713	relatedType	door
	IfcDoor	#63713	relatedName	M_Single-Flush:0915 × 2134 mm:184,917
	IfcMaterialList	#63755	finMaterial	Door Frame/Panel
	IfcSpaceBoundary	#160914	relatedID	#104786
	IfcWallStandardcase	#104786	relatedType	wall
	IfcWallStandardcase	#104786	relatedName	Basic Wall:Interior - Partition (92 mm Stud):224,960
	IfcMaterialayerSet	#26383	finMaterial	Plasterboard
	IfcSpaceBoundary	#160915	relatedID	#34382
	IfcWallStandardcase	#34382	relatedType	wall
	IfcWallStandardcase	#34382	relatedName	Basic Wall:Interior - Furring (38 mm Stud):151,136
	IfcMaterialayerSet	#25964	finMaterial	Plasterboard
	IfcSpaceBoundary	#160916	relatedID	#34382

4) IFC object inference

To retrieve the related work records that will be referenced for a maintenance work, in this case, we use the Simple Protocol and RDF Query Language (SPARQL), which was developed by the W3C RDF Data Access Working Group [27], to show the list of related records for a given IFC object. SPARQL retrieves the maintenance records stored in the knowledge base, which related the specific IFC object with a set of datatype and object properties as a knowledge by RDF structure. The SPARQL is expressed as a list of conditions in a triple structure, similar to the structure of knowledge in the inferred knowledge from the reasoning engine by XML to retrieve the knowledge by the specific conditions.

4.2.1. Maintenance

Traditionally, in the process of maintenance, the work records related to a given element can only be searched to a limited extent, and the accuracy and quantity of the historical work records cannot be ensured. Since the proposed ontology integrates the inferred knowledge of the records and the element, related work records can be identified by querying the integrated FM information. Table 5 features a list of 12 maintenance work records related to the space named '139', a meeting room, from the total 590 historical maintenance work records. In this regard, we could also recognize the seven walls as space boundaries by IfcWallStandardcase entities. In addition, the seven walls have two types of finishing materials—plasterboard with four walls (#31714, #31982, #31893, #31804) and paint with three walls (#45712, #45623, #45534). In this case, the maintenance work records contained the related space. However, since facility managers could not

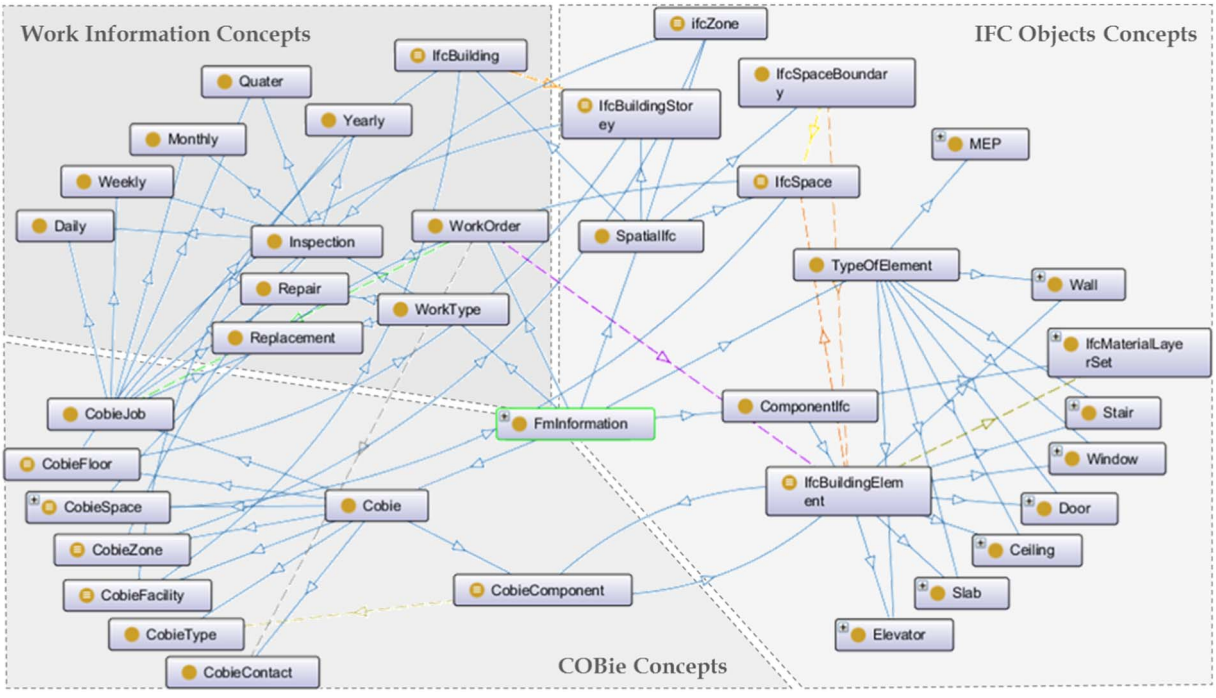


Fig. 10. Object properties in the developed ontology.

Table 5
Related maintenance work records of the space '139'.

Area	Elements	Date	Object type	IFC objects	Work contents
Door	Access door	20160602	IfcDoor	#66803	Repair the damaged glass door in meeting room
Curtain	Curtain	20160530	IfcSpace	#8721	Repair the line of blind in meeting room
Tile	Ceiling tile	20160425	IfcCovering	#160976	Repair the Cross T-bar of ceiling tiles in meeting room
Architecture	Common	20160404	IfcSpace	#8721	Repair the damaged part of beam projector connection and blind in meeting room
Door	Door & Lock-set	20160323	IfcDoor	#66803	Repair the noise and handle of the door in meeting room
Paint	Water paint	20160315	IfcWallStandardcase	#45712, #45623, #45534,	Sealing work of walls in meeting room and common area
Architecture	Common	20160314	IfcWallStandardcase	#31714, #31982, #31893, #31804	Putty work of the crack on the wall in meeting room
Architecture	Common	20160202	IfcSpace	#8721	Check and repair the utilities in meeting room
Architecture	Common	20160530	IfcSpace	#8721	Attach the white board in meeting room
Paint	Water paint	20160412	IfcWallStandardcase	#45712, #45623, #45534	Sealing work in meeting rooms at night
Tile	Ceiling tile	20160204	IfcCovering	#66803	Repair the ceiling tile in meeting room
Paint	Oil paint	20160115	IfcWallStandardcase	#31714, #31982, #31893, #45712, #45623, #45534, #31804	Painting work after repair the damaged walls in meeting room

Table 6
Shared objects in adjacent spaces.

Element name	Object type	Space	Date	Work contents
Basic Wall:Interior - Partition (92 mm Stud):148,942	IfcWallStandardcase	139	20160314	Putty work of the crack on the wall in meeting room
			20160115	Painting work after repair the damaged walls in meeting room
		140	20160510	Attach the white board on the wall in office room
			20160425	Putty and painting work in the damaged wall in office room
M_Single-Flush:0915 × 2134 mm:186,884	IfcDoor	139	20160602	Repair the damaged glass door in meeting room
			20160323	Repair the noise and handle of the door in meeting room
		140	20160502	Repair the damaged part at the front of glass door in office room
			20160411	Check and repair the doors in office room
			20160325	Repair the baseboard at the front of door in office room

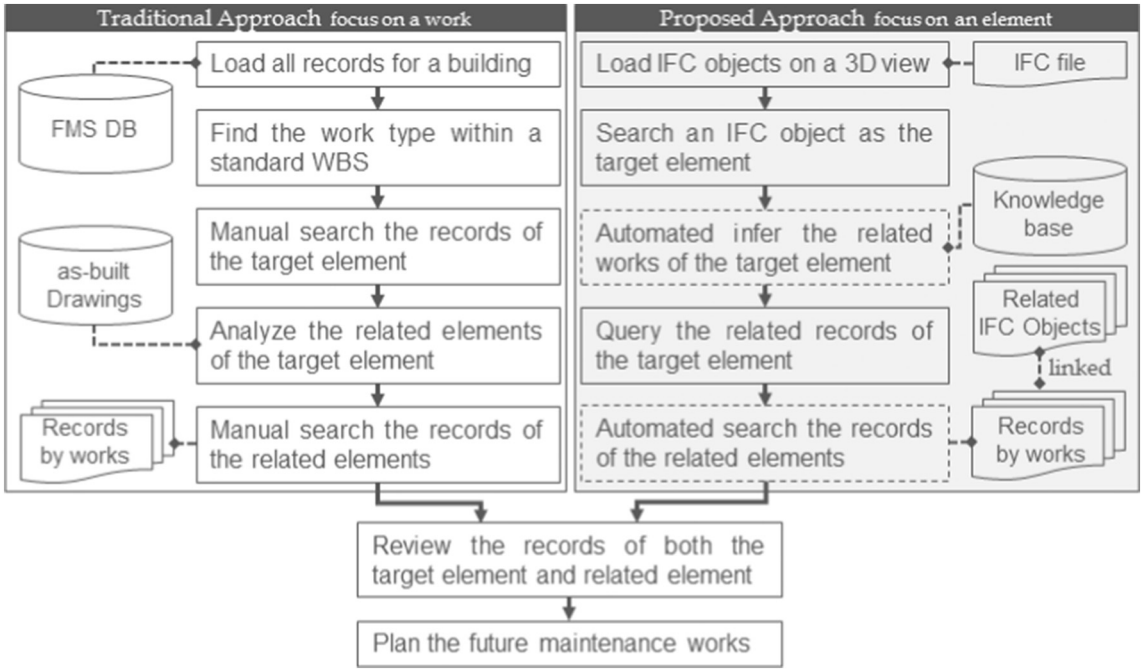


Fig. 11. Comparison of traditional and proposed approach.

input the data, many FMS DBs contain no space information regarding maintenance work on site. For this reason, the data management of the FM work could be improved by adapting to BIM in FM practice.

4.2.2. Operation

Moreover, to operate a facility, FM information managed focusing on the work (e.g., the resource, duration, and level of difficulty of the FM work) in the current FM process. However, since the proposed ontology provides object-oriented inferred knowledge, the FM information can be managed based on IFC objects to increase the availability from a facility operation, such as space management, energy monitoring, or interior remodeling. Using the query of inferred knowledge, the historical work records have been conducted in other spaces, which share an IFC object. Typically, since spaces share an IFC object, such as walls and doors in the case of adjacent spaces, the work records should be managed together when FM works have been

conducted on the same object but in different spaces. In this case, the maintenance work records should be semantically included in the list of records of adjacent spaces. In this case study, space 139 is located next to space 140, which is an office space with two shared IFC objects—a wall (#31982) and a door (#66803). Table 6 shows the case where adjacent spaces share a wall and door. In such cases, the BIM information is assumed by the as-is BIM model. However, BIM models are seldom updated in FM practice, in terms of the hardware required to run the BIM authority software. In addition, most facility managers lack the skill and experience to manage a BIM model. Therefore, to operate based on BIM using the Semantic Web, the facility manager should be trained in both BIM and the Semantic Web.

4.2.3. FM work management

When facility managers develop a plan for the maintenance work in a building, they currently use a traditional approach that focuses on text-based work records. In this regard, the manager has to find the work type from a work breakdown structure of FM works in FMS without considering the target object. Next, the manager must manually search for records of the target element and related elements. In the proposed approach, however, a facility manager could use linked BIM data using the ontological knowledge base with FMS DB. When the manager defines the target object for maintenance, the related elements are also defined according to the IFC schema, such as space boundaries. Since the ontology provides connectivity among the IFC objects and work records, the manager can obtain integrated information, thereby ensuring the consistency and accuracy of the results. Moreover, if the manager analyzes every element of a building for maintenance in the future, the traditional approach cannot support the result, but only work types can be used to analyze the frequency and effectiveness (Fig. 11).

Using the proposed ontology, the relative building elements of an FM work are automatically recognized by object type and space information using IFC schema. Thus, a facility manager can plan FM work scheduling by BIM, and the work scope can be identified based on objects linked to work information. If an FM work is conducted in multiple spaces with many IFC objects, then scheduling becomes critical for efficient implementation. When a manager applies 3D-BIM visualization, the status of multiple FM works can be managed with the detailed properties of the IFC objects, such as the material specifications referenced in the building element and the interference of FM works in the workplace (Fig. 12).

5. Discussion

In the field of construction, there is growing interest in data exchange among various stakeholders who use different tools to manage building information. In this regard, the IFC could be a solution as an international standard file format to ensure interoperability. In each phase of a building's lifecycle, the building information is generated according to the level of development (LOD) by the progress of the project. To that end, the American Institute of Architects (AIA) published the LOD specifications in 2013 to define the representation details of a BIM model during a building's lifecycle. Because the LOD is a recommendation, the consistency and capacity of the BIM model generated in design and construction phases are inadequate. Ideally, the BIM model should be a single integrated model throughout a building's lifecycle; thus, the LOD is secured in generating an IFC-based BIM model to use it in the O&M phase.

At the start of the O&M phase, the organization should develop the DB of FMS using the facility information handed over by the construction manager. In this regard, manually developing FMS DB is the best way to increase the quality of FMS in terms of usability in the O&M phase. When the facility information is handed over, the information cannot be directly entered into the FMS DB because it was created from the perspective of construction, and the objective and functions of information management differ from those of the construction phase. Even if facility information has been generated by an IFC-based BIM model as an international standard, various FMSs have different data structures with a variety of sub-modules; thus, it is necessary to customize the facility information. In the case of the commercial BIM-based FMS, the DB is separated into IFC and FMS to support the functions of the FMS. For this reason, to increase the usability of an IFC-based BIM model in FMS, an IFC object is needed for the work information of the FMS DB using ontology for a semantic data link.

The proposed ontology provides a link to IFC objects and work information in order to help search historical FM work records related to a current job. In the traditional FM work process, when a defect occurs in a building element, after a worker visits the site and checks the defect, the worker plans the maintenance with facility information. The

Date	Area	Element	Contents	Spaces	IFC Objects
20160322	Architecture	Common	Urethane Coating work in water leakage part	'155'	#28202, #28331, #28420, #28598
				'156'	#28202, #28598, #28420, #24403
				'157'	#24403, #28598, #28331, #28594

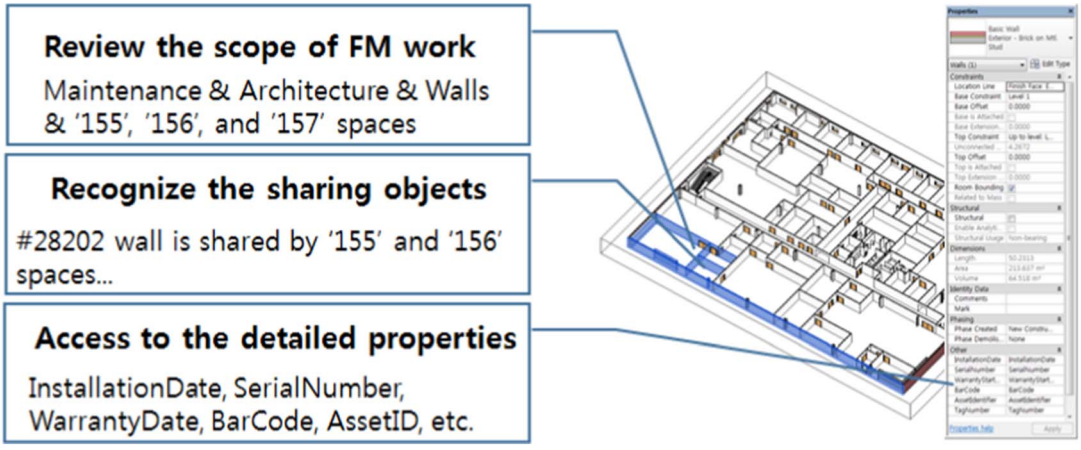


Fig. 12. Improvements of the proposed approach.

facility information, however, is comprised of 2D-based documents, separated according to information type, such as floor plans, partial details, and work records. In this regard, because the 2D-based documents cannot fully represent the parts of a building, crucial facility information is often be omitted. In addition, historical work records mainly contain text data, which can only be searched by keywords, so search results are poor in terms of quality and quantity. To that end, using the proposed ontology, semantic relationships provide the work records related not only to the space but also to the building element in order to increase the quality of the results by searching for related work records. In addition, in the case of a building element, such as a wall or door, maintenance works are usually conducted at a building element in two adjacent spaces at a similar frequency, which increases the resulting quantity of work records by about 200%.

6. Conclusion

FM is the integrated management of building premises and the services required to accommodate and support the core business activities of the major client organization. To efficiently conduct FM-related activities, some required information must be clearly defined in order to systemically manage activities during the operation and maintenance phase. To that end, various information management technologies have been developed for the FM area, such as FMS. Currently, however, most FM work is done manually. Thus, there is ample room for improvement in FM work processes through BIM-based automatic information management process that use computer-aided FMS.

Within this context, there is also growing interest in BIM in the industries of architecture, engineering, and construction, along with increased awareness about communicating information over the building's lifecycle. Using BIM-based information management, however, raises several potential problems for FM. In general, because the FMS database structure differs from BIM-based information, an inexperienced facility manager would find it difficult to utilize. BIM data are organized by object, but work information in an FMS database is based on work activity. Therefore, object-based BIM data cannot be fully used to support the work information in FMS.

This research proposes a method for integrated information management of a BIM-based FMS with relevant work information obtained via the Semantic Web. The data structure of the BIM-based mandatory FM information was analyzed and then defined as ontology in order to semantically link the IFC objects with the FM work information. To evaluate the proposed ontology, a case study was conducted using historical maintenance work records in the architecture area. Through the case study, the IFC objects were linked to historical maintenance work records, and the ontology can search semantically related historical maintenance records based on an IFC object.

This study makes several contributions: 1) the traditional FM works process can be improved by focusing on a building element as an object oriented data management; 2) the proposed approach could be a guideline for FM activity using BIM to ensure the accuracy and consistency with traditional FMS; and 3) since the proposed approach provided an object-oriented work process, the FM strategy could be improved by considering detailed and specific management of each building element.

The proposed approach will enable facility managers to use a BIM-based FMS with semantically linked data in the Semantic Web during the O&M phase in order to provide a BIM environment without the specific BIM authoring application. Additionally, the proposed process helps ensure that the data input required for facility management is captured accurately and reliably through computerized data management. This creates the opportunity to improve the economics, accuracy, and productivity of managing FM works using BIM with the Semantic Web from a practical perspective.

Future research should pursue further validation of the BIM model handed over from the construction phase, including the COBie data sheet. In addition, the proposed approach should be developed further to support various other functions of FMS from an asset management perspective. Lastly, in terms of using BIM-based facility information in an FMS, a methodology for updating the actual conditions of an existing building should be developed with consideration of linkage with a heterogonous DB.

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